

## 8.2.11

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# Question

Find the roots of the equation  $x^2 + 3x - 10 = 0$

# Theoretical Solution

We express the equation as a parabola  $y = x^2 + 3x - 10$ , which can be written in the standard conic form  $\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0$  as:

$$x^2 + 3x - y - 10 = 0 \quad (1)$$

The components of the conic form are:

$$\mathbf{V} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad 2\mathbf{u} = \begin{pmatrix} 3 \\ -1 \end{pmatrix} \implies \mathbf{u} = \begin{pmatrix} 3/2 \\ -1/2 \end{pmatrix}, \quad f = -10 \quad (2)$$

# Theoretical Solution

We find the intersection with the x-axis, which has the parametric form

$\mathbf{x} = \mathbf{h} + \kappa \mathbf{m}$ , with a point on the line  $\mathbf{h} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$  and direction vector

$$\mathbf{m} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The values of  $\kappa$  for the intersection points The solution is:

$$\kappa = \frac{-\mathbf{m}^\top (\mathbf{V}\mathbf{h} + \mathbf{u}) \pm \sqrt{(\mathbf{m}^\top (\mathbf{V}\mathbf{h} + \mathbf{u}))^2 - (\mathbf{m}^\top \mathbf{V}\mathbf{m})(\mathbf{h}^\top \mathbf{V}\mathbf{h} + 2\mathbf{u}^\top \mathbf{h} + f)}}{\mathbf{m}^\top \mathbf{V}\mathbf{m}} \quad (3)$$

We will now calculate each term in this formula.

**Denominator Term:**

$$\mathbf{m}^\top \mathbf{V} \mathbf{m} = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1 \quad (4)$$

**Numerator Term 1:** Since  $\mathbf{h} = \mathbf{0}$ , this term is  $-\mathbf{m}^\top \mathbf{u}$ .

$$-\mathbf{m}^\top \mathbf{u} = - \left( \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 3/2 \\ -1/2 \end{pmatrix} \right) = -\frac{3}{2} \quad (5)$$

# Theoretical Solution

**Term under the Square Root:** Since  $\mathbf{h} = \mathbf{0}$ , this term is  $(\mathbf{m}^\top \mathbf{u})^2 - (\mathbf{m}^\top \mathbf{V} \mathbf{m})(f)$ .

$$(\mathbf{m}^\top \mathbf{u})^2 = \left(\frac{3}{2}\right)^2 = \frac{9}{4} \quad (6)$$

$$(\mathbf{m}^\top \mathbf{V} \mathbf{m})(f) = (1)(-10) = -10 \quad (7)$$

$$\text{under root term} = \frac{9}{4} - (-10) = \frac{9}{4} + \frac{40}{4} = \frac{49}{4} \quad (8)$$

Now, we substitute these calculated values back into the formula for  $\kappa$ :

# Theoretical Solution

$$\kappa = \frac{-3/2 \pm \sqrt{49/4}}{1} = -\frac{3}{2} \pm \frac{7}{2} \quad (9)$$

This gives two values for  $\kappa$ :

$$\kappa_1 = -\frac{3}{2} + \frac{7}{2} = \frac{4}{2} = 2 \quad (10)$$

$$\kappa_2 = -\frac{3}{2} - \frac{7}{2} = -\frac{10}{2} = -5 \quad (11)$$

# Theoretical Solution

The points of intersection are found using  $\mathbf{x}_i = \mathbf{h} + \kappa_i \mathbf{m}$ :

$$\mathbf{x}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + 2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \quad (12)$$

$$\mathbf{x}_2 = \begin{pmatrix} 0 \\ 0 \end{pmatrix} - 5 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -5 \\ 0 \end{pmatrix} \quad (13)$$

The roots of the original equation are the x-coordinates of these intersection points.

**The roots are 2 and -5.**



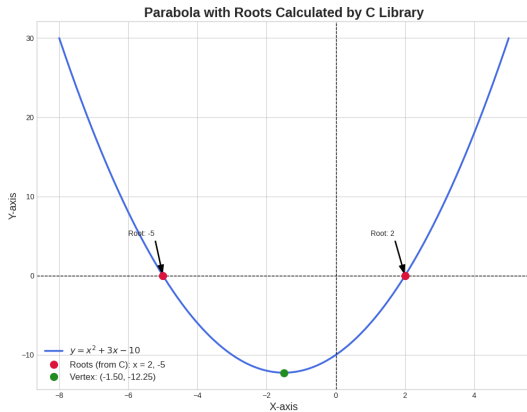


Figure: